

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1110

COURSE OUTCOME COVERED

CO4: Implement object-oriented system layers using design principles and patterns, and integrate access and view layers with OODBMS (*BL3*)
Assessment Tool Weightage: 30%

QUESTIONS: 15

Total Marks:30

Annexure A

CLASS TEST

Code of Conduct:

I Bheeshma L / 192511332 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

BHEESHMA L

RUBRICS FOR CALCULATION:

CLASS TEST

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Annexure A

CLASS TEST

QUESTIONS:

Total Marks: 30

Q1. Define architectural patterns in object-oriented systems. List any two common patterns. Explain how the Model-View-Controller (MVC) architecture improves modularity and maintainability.

[L1, L2, L3]

Q2. What is the Data Access Object (DAO) Pattern? Mention two advantages. Explain how DAO separates database-related operations from application logic.

[L2, L3, L4]

Q3. What are object-oriented design guidelines? Identify any two categories of design guidelines. Explain the role of the Liskov Substitution Principle (LSP) in developing reusable software components.

[L2, L3, L4]

Q4. Define database persistence in object-oriented applications. Mention two persistence mechanisms. Explain how an Object-Relational Mapping (ORM) framework supports persistent object management.

[L1, L2, L3]

Q5. What is an Application Layer? State two major responsibilities. Explain how the Application Layer coordinates use cases without directly implementing domain business rules.

[L2, L3, L4]

Q6. Explain the structure of a three-tier architecture consisting of Presentation, Business, and Data layers. Analyze the flow of a customer transaction through these layers in an e-commerce application.

[L2, L4, L5]

Q7. What is the role of a Business Logic Layer? List two important functions. Explain how it validates and processes requests received from the presentation layer.

[L2, L3, L4]

Q8. Explain the significance of cohesion and coupling in object-oriented design. Analyze how high cohesion and low coupling contribute to software maintainability and reusability.

[L3, L4, L5]

Q9.What is encapsulation in an object-oriented system? Explain how encapsulation protects object state and compare it with data hiding using a suitable example.

[L2, L4]

Q10.Design a layered object-oriented system for an online banking application. Identify the major layers, recommend suitable design patterns, and justify how the architecture improves security, scalability, and maintainability.

[L3, L4, L5]

Q11.What is the role of the Infrastructure Layer in object-oriented architecture? List two responsibilities. Explain how it supports technical services such as database access and external system communication.

[L1, L2, L3]

Q12.Define the Singleton Design Pattern. Mention its major characteristics. Explain its appropriate use in managing shared resources and discuss one limitation of the pattern.

[L2, L3, L4]

Q13.What is dependency injection in object-oriented design? List two benefits. Analyze how dependency injection reduces tight coupling between classes and improves testability.

[L2, L3, L4]

Q14.Explain the working of the Strategy Design Pattern. Compare the roles of Context and Strategy with a suitable real-world application example.

[L3, L4, L5]

Q15.Design a layered architecture for a Library Management System. Identify suitable layers, recommend relevant design patterns, and justify how your proposed architecture supports flexibility and maintainability.

[L3, L4, L5]

20/02/26

CSA11
SLOT-D

CLASS TEST

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- CO4

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① Define architectural patterns in object-oriented systems.

⇒ An architectural pattern is a general, reusable solution that defines the high-level structure of a software system.

Two common patterns — Model-view-controller (MVC)

layered (N-tier)
Architecture

MVC separates the application into,

It shows data & business logic

handles user interface & presentation

well-defined responsibility

developers can easily modify the GUI interface of any data.

⇒ Model
⇒ View
⇒ Controller

② The Data Access Object (DAO) is a structural design pattern that provides an abstract interface to a database or encapsulating all data access operations (create, alter, delete etc..)

Two Advantages

improved maintainability

1 Promotes code reusability and testing.

The DAO implementation internally handles the connections, interfaces and data in a database system.

- ② OOAD design guidelines are set of principles / rules used to design classes, relationships, entities and functions, and interactions so that the resulting system is understandable, flexible, reusable and easy to maintain.

Two categories

class design guidelines (relationships)

Relationship / association design guidelines

(e.g. inheritance, coupling)

Role of Liskov Substitution principle

class must be substitutable for objects of the base (parent class) without altering the correctness of the program / logic.

- ④ Data persistence is the ability of the object oriented application to store the data of the objects in permanent storage (database), so that the data survives beyond the program's execution and can also be retrieved later when needed.

Two persistence mechanisms

ORM - Object
Relational mapping

Serialization
to files or (OOBMS)

An ORM - Object Relational mapping automatically maps object classes, attributes to their respective rows and columns. It also generates the necessary SQL data or query for processing the object.

- ⑤ An application layer is the layer of the software architecture that defines and orchestrates the job, the software is the top layer of the architecture.

It acts as a thin coordinate layer between the presentation layer & the Domain layer.

Two major Responsibilities



Coordinating & sequencing
calls to domain objects



Managing transactions,
security, application
work flow.

The Application layer itself does not contain the domain logic, it simply directs the flow, receiving request → invoking the relevant domain objects in the order of the application work flow.